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ORIGINAL PAPER

# Anti-obesity activity of Yamabushitake (*Hericium erinaceus*) powder in ovariectomized mice, and its potentially active compounds

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**Abstract** *Hericium erinaceus* (*H. erinaceus*) improves the symptoms of menopause. In this study, using ovariectomized mice as a model of menopause, we investigated the anti-obesity effect of this mushroom in menopause. Mice fed diets containing *H. erinaceus* powder showed significant decreases in the amounts of fat tissue, plasma levels of total cholesterol, and leptin. To determine the mechanism, groups of mice were respectively fed a diet containing *H. erinaceus* powder, a diet containing ethanol extract of *H. erinaceus*, and a diet containing a residue of the extract. As a result, *H. erinaceus* powder was found to increase fecal lipid levels in excreted matter. Further in vitro investigation showed that ethanol extract inhibited the activity of lipase, and four lipase-inhibitory compounds were isolated from the extract: hericenone C, hericenone D, hericenone F, and hericenone G. In short, we suggest that *H. erinaceus* has an anti-obesity effect during menopause because it decreases the ability to absorb lipids.

**Keywords** *Hericium erinaceus* · Anti-obesity · Menopause · Pancreatic lipase inhibition · Hericenone

E. Hiraki and S. Furuta equally contributed to this work.

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## Introduction

Menopause is an unavoidable process that usually affects women between the ages of 40 and 60 years. After 12 months of permanent loss of menstruation, a woman is considered to be in menopause [1]. Menopause is characterized by falling levels of estrogen and progesterone, which can lead to the development of symptoms including hot flashes, night sweats, vaginal dryness, mood swings, declining libido, insomnia, lethargy/fatigue, irritability, anxiety, depression, heart palpitations, and joint pain [2]. Hormone replacement therapy has been used as an effective method of treating various menopause-related symptoms [3]; however, the Women's Health Initiative study found that treating women for greater than 5 years with estrogen plus progestin led to a significant increase in the risk for breast cancer and other diseases [4, 5]. Consequently, it is necessary to develop natural treatments for menopause with fewer undesirable side effects.

Obesity is one of the symptoms of menopause, and it can be very problematic for women's health [6] since it can lead to excessive fat-related metabolic diseases and chronic diseases, including type 2 diabetes mellitus, hypertension, and dyslipidemia [7–9]. In addition, excess weight is generally linked to the onset of several major chronic diseases such as cardiovascular diseases and cancers [10]. There is also strong evidence that obesity is associated with increased morbidity

and mortality. Therefore, it is essential to investigate solutions to the problem of obesity in menopause.

*Hericium erinaceus* (*H. erinaceus*) has been eaten in East Asia for a very long time, and it has many pharmacological effects, including antitumor [11, 12], antimicrobial [13], antioxidant [14], and immunomodulatory activities [15], the synthesis of nerve growth factor [16], and the improvement of mild cognitive impairment [17]. Our research group has shown that *H. erinaceus* improves depression and anxiety in menopausal women in clinical practice [18]. Although an anti-obesity effect of *H. erinaceus* in obesity model mice has been reported [19], we focus in this study on obesity in menopause. Using ovariectomized mice as a model of menopause, the fat-reducing effects of *H. erinaceus* and its active compounds were revealed.

## Materials and methods

### Materials

4-Methylumbelliferyl oleate was obtained from Sigma (St. Louis, MO, USA), and orlistat was obtained from Cayman Chemical Co. (Ann Arbor, MI, USA). Nembutal was purchased from Dainippon Sumitomo Pharma Co., Ltd. (Osaka, Japan). *d*-Chloroform was purchased from Kanto Chemical Co., Inc. (Tokyo, Japan). Tris(hydroxymethyl)aminomethane was purchased from Tokyo Chemical Industry Co., Ltd. (Tokyo, Japan). Unless otherwise specified, chemicals and organic solvents were obtained from Wako Pure Chemical Industries Co. (Osaka, Japan).

### In vivo validation

#### *Extraction of H. erinaceus*

Fruiting bodies of *H. erinaceus* were obtained from Aso Biotech Co. (Kumamoto, Japan). Freeze-dried fruiting bodies of *H. erinaceus* were ground and extracted with ethanol at room temperature. The solution was evaporated under vacuum to afford ethanol extract and a residue of the extract (yield: 24.55 and 72.13%, respectively). This residue was suspended in MilliQ water and centrifuged at 3,000 rpm for 10 min, and the supernatant was freeze-dried to afford the water extract (yield: 23.77%).

#### *Animals*

All experiments were reviewed by the animal experiment ethics committee of Kyushu University (A22-234-2) and conducted after approval of the study protocol. Ovariectomized C57BL/6JJcl mice (12 weeks old, female), which are a model of menopause, were purchased from CLEA Japan Co. (Tokyo, Japan). They had free access to water and diet, and were maintained on a 12-h light/dark cycle (light on at 8:00 and off at 20:00) at 23 ± 2 °C. AIN-76 diets were used on the basis of the diet reported by the American Institute of Nutrition Ad Hoc Committee on Standards for Nutritional Studies. A modified AIN-76 diet based on the energy intake ratio and PFC (protein, fat, and carbohydrate) balance of Japanese menopausal women in 2008 was used as the control diet (Table 1) [20].

#### *Study 1: effects on fat reduction*

*Animals and diets* After randomization, the mice were placed in either the control diet group (CD, $n = 12$) or the 5% fruiting body (w/w) of *H. erinaceus*-containing diet group (FBD, $n = 12$) for 16 days. Based on our previous study regarding mushrooms such as *Ganoderma lingzhi* and *Pleurotus eryngii*, we predicted that a dose of 5% fruiting body would not cause diarrhea in animal experiments. A dose of >5% may cause diarrhea because it contains too much dietary fiber. Because the aim of this study was to evaluate the potential of *H. erinaceus*, we selected a dose of 5% fruiting body as the highest dose that does not cause adverse effects. The final dose of *H. erinaceus* powder itself in the FBD was about 0.09 g/day because the average food intake in the FBD group was 1.87 ± 0.10 g/day. The energy composition of *H. erinaceus* was analyzed at the TOYO Environmental Laboratory (Miyazaki, Japan) (Table S1 in the Electronic supplementary material, ESM), and the PFC balances between CD and FBD were equalized. The compositions of the experimental diets are presented in Table 1. The energy intake of all of the mice was adjusted by pair feeding.

**Table 1** Compositions of the experimental diets (study 1)

| Ingredients (g/100 g) | CD | FBD |
|---|---|---|
| Sample | – | 5 |
| Vitamin mix | 1 | 1 |
| Mineral mix | 3.5 | 3.5 |
| DL-Methionine | 0.3 | 0.3 |
| Choline bitartrate | 0.2 | 0.2 |
| Cellulose | 5 | 2.73 |
| Sunflower oil | 12.71 | 12.44 |
| Corn starch | 15.49 | 15.39 |
| Sucrose | 41.15 | 39.71 |

*Body fat* At the end of the experimental period, the mice were anesthetized with intraperitoneal administration of Nembutal, and the amount of white adipose tissue in visceral and subcutaneous fat was measured by X-ray computed tomography (X-ray CT) (Latheta LCT-100, Hitachi Aloka Medical, Ltd., Tokyo, Japan) at a scan pitch of 2 mm for mouse body parts between the diaphragm and the base of the tail.

*Amount of tissues and plasma parameters* The day after the body fat was measured, anesthetized mice were killed by decapitation after 4 h fasting. Blood was collected from the heart of each mouse, and then the liver, uterus, kidney, brain, and adipose tissues (around the kidney and uterus) were excised. Plasma triglyceride, total cholesterol, and glucose were determined by the TG E test, cholesterol E test, and glucose CII test (Wako, Japan), respectively. Leptin and adiponectin levels were determined by ELISA kits (BD Bioscience, San Jose, CA, USA). All kits were used in accordance with the manufacturer's protocols.

*Study 2: effects on lipid excretion*

*Animals and diets* After randomization, the mice were divided into four groups ($n = 6$) that were respectively fed the CD, FBD, ethanol extract of *H. erinaceus*-containing diet (EED), or a residue of the extract-containing diet (RD) for 48 days. We set the doses of EED and RD based on the amounts of ethanol extract (0.86%) and residue (4.15%) obtained from the amount (5%) of fruiting body of *H. erinaceus* contained in FBD. The energy compositions of *H. erinaceus* and the residue were analyzed at Kumamoto Industrial Research Institute (Kumamoto, Japan) (Table S2 in the ESM), and that of the ethanol extract was calculated by subtracting that of the residue from that of *H. erinaceus*. The PFC balances among the four groups were then equalized. The compositions of the experimental diets are presented in Table 2. The energy intake of each mouse was adjusted by pair feeding.

*Body fat* On days 19 and 42, the amount of white adipose tissue in the visceral and subcutaneous fat was measured by X-ray CT using the same methods used in study 1.

*Amount of tissues* At the end of the experimental period, mice were killed by decapitation after 4 h of fasting, and the liver, adipose tissues (around the venter, kidney, and uterus), soleus muscle, gastrocnemius muscle, and plantaris muscle were excised.

*Fecal lipid excretion* During the experimental period from day 8 to day 14, excreted stools were collected and the lipid level of the excreted material was measured using the method of Folch, as previously described [21]. In brief, 500 mg of each excretion were dissolved in threefold the volume of solvent [chloroform:methanol = 2:1 (v/v)]. After 1 h of shaking, it was centrifuged at $1.2 \times 10^3$ rpm for 10 min and the supernatant was then obtained. This process was conducted two times. To remove salts and other water-soluble materials, 0.75% potassium chloride (20%, v/v) was added. The chloroform–methanol layer was obtained and evaporated under a vacuum and a nitrogen gas stream, and the extracted material was regarded as the lipid present in the matter excreted by the mice.

**Table 2** Compositions of the experimental diets (study 2)

| Ingredients (g/100 g) | CD | FBD | EED | RD |
|---|---|---|---|---|
| Sample | – | 5 | 0.86 | 4.15 |
| Vitamin mix | 1 | 1 | 1 | 1 |
| Mineral mix | 3.5 | 3.5 | 3.5 | 3.5 |
| DL-Methionine | 0.3 | 0.3 | 0.3 | 0.3 |
| Choline bitartrate | 0.2 | 0.2 | 0.2 | 0.2 |
| Cellulose | 0 | 0 | 0 | 0 |
| Casein | 20.65 | 20.65 | 20.65 | 20.65 |
| Sunflower oil | 3.29 | 3.29 | 3.29 | 3.29 |
| Lard | 9.41 | 9.41 | 9.41 | 9.41 |
| Corn starch | 15.49 | 14.13 | 15.26 | 14.36 |
| Sucrose | 41.15 | 37.52 | 40.53 | 38.14 |

## In vitro validation

*Lipase-inhibitory activity*

The pancreatic lipase inhibition activities of the extracts were measured by an in vitro enzyme reaction in which 4-methylumbelliferyl oleate (4-MUO) was used as the substrate [22]. The extracts dissolved in 3 µl of DMSO were first diluted with 25 µl buffer consisting of 13 mM Tris–HCl, 150 mM NaCl, and 1.3 mM $CaCl_2$ (pH 8.0). The sample solution was then mixed with 50 µl of a 0.25 mM 4-MU solution (dissolved in 2 µl DMSO and the above buffer) in the well of a 96-well plate, and 25 µl of the lipase solution (50 U/ml) were added. After incubation at 25 °C for 30 min, 100 µl of 0.1 M sodium citrate (pH 4.2) were added to stop the enzyme reaction. The amount of 4-methylumbelliferone released by lipase was measured with a fluorometric microplate reader (FlexStation 3 Microplate Reader, Molecular Devices, Sunnyvale, CA, USA) at an excitation wavelength of 355 nm and an emission wavelength of 460 nm. DMSO and orlistat (final concentration: 1 µM) were used as a negative control and positive control, respectively.

*Isolation of active compounds*

The isolation of active compounds from ethanol extract of the fruiting body of *H. erinaceus* was conducted based on lipase-inhibitory-activity-guided fractionation. Ethanol extract (110 g) was extracted with *n*-hexane, and the *n*-hexane fraction (6.629 g) and the residue fraction were obtained. Similarly, the residue of ethanol extract was extracted with water and the water extract was obtained. The *n*-hexane fraction components were separated by silica gel column chromatography (17 cm i.d. × 60 cm, 600 g of Wakogel C-200) and eluted with an *n*-hexane/EtOAc solvent system [10:0 (2 l), 9.7:0.3 (2 l), 9.6:0.4 (2 l), 9.5:0.5 (2 l), 9.4:0.6 (2 l), 9.3:0.7 (2 l), 9.2:0.8 (2 l), 9.1:0.9 (2 l), 9:1 (2 l), 7.5:2.5 (2 l), 5:5 (2 l), 2.5:7.5 (2 l), 0:10 (2 l)] and an EtOAc/MeOH solvent system [5:5 (2 l), 0:10 (4 l)]. Based on the TLC results, those fractions were combined into 11 fractions [Fr. 1 (329.8 mg), Fr. 2 (223.7 mg), Fr. 3 (547 mg), Fr. 4 (1649.4 mg), Fr. 5 (372.6 mg), Fr. 6 (108.4 mg), Fr. 7 (695.9 mg), Fr. 8 (559.5 mg), and Fr. 9 (398.8 mg)]. Then the Fr. 4 (637.1 mg) fraction was subjected to silica gel column chromatography (5.0 cm i.d. × 50 cm, 150 g of Wakogel C-200) and eluted with an *n*-hexane/chloroform solvent system [10:0 (0.8 l), 9:1 (0.8 l), 8:2 (0.8 l), 7:3 (0.8 l), 6:4 (0.8 l), 5:5 (0.8 l), 4:6 (0.8 l), 3:7 (0.8 l), 2:8 (0.8 l), 1:9 (0.8 l), 0:10 (0.8 l)] and methanol 100% (2 l). The obtained subfractions were combined into three fractions [Fr. 4-1 (5.95 mg), Fr. 4-2 (517.91 mg), Fr. 4-3 (53.54 mg)]. The Fr. 4-2 fraction was separated by reverse-phase preparative high-performance liquid chromatography (HPLC) [column: Inertsil ODS-3 20 × 250 mm (GL Sciences, Tokyo, Japan), mobile phase: isocratic MeOH, 8.0 ml/min, UV detection: 238 nm, 279 nm], which afforded compounds **1** (15.73 mg), **2** (6.66 mg), **3** (2.46 mg), and **4** (24.62 mg).

*NMR and MS spectrometry*

Compounds **1**, **2**, **3**, and **4** were analyzed by 400 MHz nuclear magnetic resonance (NMR). The samples were dissolved in chloroform-*d*, and the internal reference standard was tetramethylsilane (TMS). $^{1}$H and $^{13}$C NMR and DEPT spectra were collected with standard pulse sequences. FAB-MS data were obtained using a JMS-700 mass spectrometer (JEOL Ltd., Tokyo, Japan).

*Quantitation of the constituents of the extracts*

The separation and quantitation of the constituents of the extracts were accomplished using liquid chromatography (Agilent 1220 Infinity LC, Agilent Technologies Japan, Ltd., Tokyo, Japan). The stationary phase was an Inertsil ODS-3 column (4.6 × 150 mm, 5 μm) from GL Sciences Inc. The mobile phase consisted of methanol at a flow rate of 1 ml/min. The elution mode was isocratic. The injection volume was 5 μl, and the temperature for the analytical column was set at 40 °C. The quantitation of constituents in extracts was accomplished using a UV–vis detector. The detection wavelength was 295 nm.

### Statistical analysis

Almost all data were expressed as the mean ± SEM except the data for the lipase activity, which were expressed as the mean ± SD. Statistical significance was determined using Student's *t* test for study 1. One-way analysis of variance followed by the Tukey–Kramer test were conducted for study 2. Dunnett's test was conducted on lipase activity assays. A $P$ value of either <0.05 or <0.01 was considered significant. Logarithmic transformation was used to normalize data with a skewed distribution before statistical analysis.

## Results

### In vivo validation

*Study 1*

*Body fat* In study 1, the mice were fed with CD or FBD for 14 days. The final dose of *H. erinaceus* powder itself in the FBD was about 0.09 g/day. Although there was no significant difference in body weight between the two groups, on the last day, the FBD group showed a significant reduction in white adipose tissue in the visceral and subcutaneous fat compared with the CD group (Fig. 1).

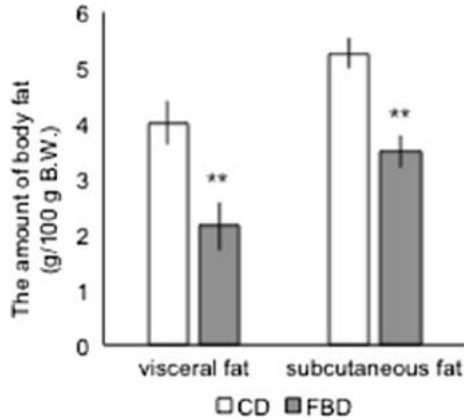


**Fig. 1** Amount of body fat after 14 days of feeding in study 1. ** indicates a significant difference compared with the CD group ($p < 0.01$, Student's *t* test). $n = 12$. Mean ± SEM

*Amounts of tissues and plasma parameters* The FBD group showed a decrease in the amount of fat tissue around the uterus and the kidney compared with the CD group (Table 3). In terms of the plasma parameters, the levels of total cholesterol and leptin were also significantly decreased in the FBD group. Additionally, the amount of adiponectin was increased (per the amount of fat tissue).

*Study 2*

*Body fat* In study 2, each mouse got all four diets for 42 days. The body weights of the mice did not change in any of the four groups (Fig. S1 in the ESM). Only the CD group had gained significant visceral fat as of day 19 (Fig. 2). By day 42, the EED group also gained the same amount of visceral fat as the CD group; however, the FBD and RD groups maintained their fat levels from the first day, and both groups showed significantly reduced visceral fat compared to the CD group on day 42. The EED group had a significantly higher amount of visceral fat than the FBD group on day 42.

As for subcutaneous fat, a significant increase was confirmed in the CD and EED groups on day 19, although the FBD and RD groups did not show such an increase. On day 42, the amount of subcutaneous fat in the EED and RD groups increased by the same amount as it did in the CD group. On the other hand, the FBD group maintained the same level of fat as that present on the first day, and showed a significantly lower amount of subcutaneous fat than the CD group on day 42. The EED group had a significantly higher amount of subcutaneous fat than the FBD group on day 42.

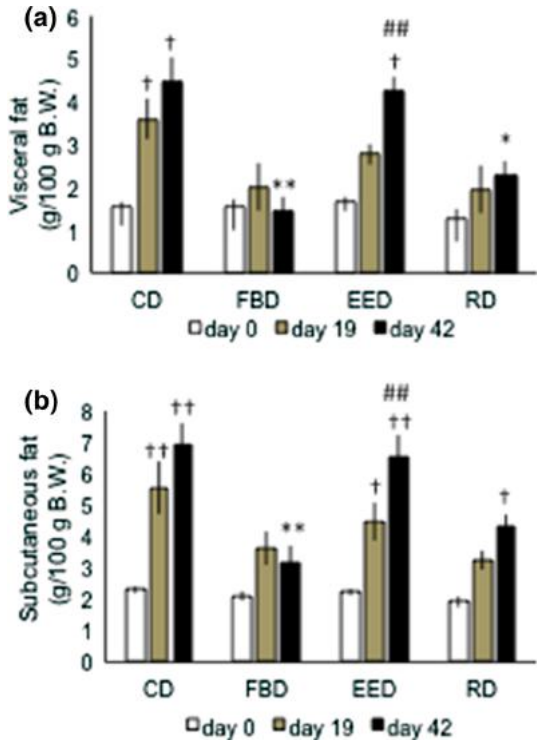


**Fig. 2** The amount of body fat in study 2. **a** Visceral fat and **b** subcutaneous fat. *Double dagger* and *single dagger* indicate significant differences from the amount of fat on the first day in each group ($p < 0.01$ and $p < 0.05$, respectively, Tukey–Kramer test). ** and * indicate significant differences from the amount of fat of the CD group ($p < 0.01$ and $p < 0.05$, respectively, Tukey–Kramer test). ## indicates a significant difference from the amount of fat of the FBD group ($p < 0.01$, Tukey–Kramer test). $n = 6$. Mean ± SEM

**Table 3** Amounts of tissues in study 1

| | CD | | FBD | |
|---|---|---|---|---|
| | Mean | SEM | Mean | SEM |
| Body weight (g) | 22.2 | 0.4 | 21.8 | 0.3 |
| Organ weight (mg) | | | | |
| Liver | 994 | 40 | 996 | 66 |
| Uterus | 12.2 | 1.5 | 10.3 | 4.3 |
| Kidney | 233 | 6 | 252 | 10 |
| Brain | 441 | 7 | 446 | 7 |
| White adipose tissue | | | | |
| Perirenal | | | | |
| (mg) | 181 | 19 | 98.4** | 9.0 |
| (mg/100g B.W.) | 806 | 79 | 451** | 39 |
| Around uterus | | | | |
| (mg) | 425 | 43 | 257** | 23 |
| (mg/100g B.W.) | 1883 | 170 | 1180** | 104 |
| Plasma parameter | | | | |
| Triglyceride (mg/dL) | 93.1 | 8.8 | 77.3 | 2.7 |
| Total cholesterol (mg/dL) | 112 | 8 | 77.7** | 7.1 |
| Glucose (mg/dL) | 445 | 33 | 441 | 30 |
| Leptin (ng/mL) | 27.6 | 4.5 | 13.1** | 2.1 |
| Adiponectin (ng/mL) | 0.192 | 0.006 | 0.184 | 0.004 |

Statistical differences were discerned using Student's *t* test. ** indicates a significant difference compared with the CD group ($p < 0.01$). $n = 12$, except for leptin (CD group: $n = 7$, FBD group: $n = 8$) and adiponectin (CD group: $n = 6$, FBD group: $n = 9$)

*Amounts of tissues* Although there was no statistical variation in the weights of the muscles and the adipose tissue around the kidney and uterus among the four groups, the adipose tissue around the venter in the FBD group was significantly decreased compared with that in the CD group (Table 4).

*Fecal lipid excretion* The amount of lipid excreted from feces was increased in the FBD group (Fig. 3), while neither the lipid levels in the diet nor the amount of excreted matter differed significantly among the four groups. Lipid excretion in the EED and RD groups was also increased, but their levels did not reach that of the FBD group.

**Table 4** Amounts of tissues in study 2

| | CD | | | FBD | | | FED | | | RD | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Mean | | SEM | Mean | | SEM | Mean | | SM | Mean | | SEM |
| Liver (mg/100 g B.W.) | 4470 | ± | 129 | 4642 | ± | 224 | 4949 | ± | 192 | 4520 | ± | 152 |
| White adipose tissue (mg/100 g B.W.) | | | | | | | | | | | | |
| Around kidney | 712 | ± | 85 | 432 | ± | 53 | 602 | ± | 92 | 487 | ± | 62 |
| Venter | 1413 | ± | 236 | 625 | ± | 120** | 1338 | ± | 224 | 1124 | ± | 205 |
| Around uterus | 358 | ± | 75 | 255 | ± | 45 | 340 | ± | 55 | 396 | ± | 127 |
| Brown adipose tissue (mg/100 g B.W.) | 270 | ± | 23 | 287 | ± | 36 | 246 | ± | 24 | 303 | ± | 39 |
| Muscle (mg/100 g B.W.) | | | | | | | | | | | | |
| Soleus | 31.8 | ± | 3.2 | 26.8 | ± | 2.5 | 26.7 | ± | 3.8 | 31.8 | ± | 3.0 |
| Plantaris | 50.4 | ± | 4.5 | 55.9 | ± | 3.6 | 52.1 | ± | 7.2 | 55.9 | ± | 6.0 |
| Gastrocnemius | 425 | ± | 60 | 408 | ± | 10 | 405 | ± | 9 | 452 | ± | 31 |

Statistical differences were discerned using the Tukey–Kramer test. ** indicates a significant difference compared with the CD group ($p < 0.01$). $n = 6$

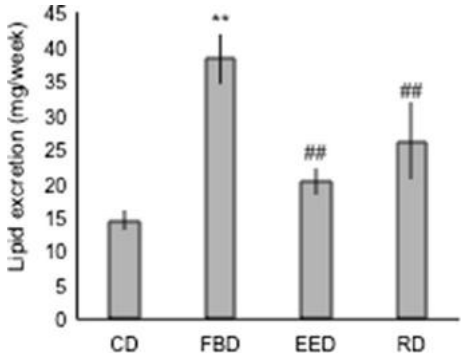


**Fig. 3** Fecal lipid excretion (mg/week). ** and ## indicate significant differences from the CD group and the FBD group, respectively ($p < 0.01$, Tukey–Kramer test). $n = 6$. Mean ± SEM

## In vitro validation

### *Lipase-inhibitory activity and compounds*

The extracts, residue, and fractions were tested for their ability to inhibit pancreatic lipase in vitro. The ethanol extract and *n*-hexane fraction caused significant inhibition of lipase. The *n*-hexane fraction showed particularly high lipase-inhibitory activity. In further bioguided fractionation, four compounds were isolated as lipase-inhibitory compounds (Fig. 4). On the basis of spectroscopic analyses, including $^{1}$H and $^{13}$C NMR spectroscopy and FAB-MS, the structures of the isolated compounds were identified as hericenone C (**1**), hericenone D (**2**) [23], hericenone F (**3**), and hericenone G (**4**) [24] (Fig. 5).

Figure 6 presents HPLC chromatograms of the ethanol extract, *n*-hexane fraction, *n*-hexane residue fraction, and water extract shown in Fig. 4a. The ethanol extract contained 18.5, 332.2, 7.3, and 7.9 μg/g of hericenones C, D, F, and G, respectively. The *n*-hexane fraction contained 23.1, 416.1, 8.4, and 8.6 μg/g of hericenones C, D, F, and G, respectively. However, peaks from the hericenones were not detected in the HPLC chromatograms of the *n*-hexane residue fraction and water extract.

## Discussion

In this study, we investigated the effect of *H. erinaceus* on the reduction of body fat in menopause. Study 1 and study 2 implied that the fruiting body of *H. erinaceus* decreased visceral and subcutaneous fat, fatty tissue around the kidney and venter, and the level of leptin. Those parameters have been found to be positively correlated with the fat cell number [25]. Therefore, eating the fruiting body of *H. erinaceus* decreases fat cells and improves body fat condition and obesity. Additionally, the FBD group showed significant decreases in total cholesterol and an increase in adiponectin; hence, *H. erinaceus* improves hyperlipidemia (a complication of obesity). Menopausal women are vulnerable to obesity because of the metabolic abnormality associated with decreased ovarian hormone production [26]. Our study demonstrated the anti-obesity activity of *H. erinaceus* using ovariectomized mice, a model of menopause. However, Hiwatashi et al. [19] reported anti-obesity activity of *H. erinaceus* in male mice fed a high-fat diet. Therefore, the anti-obesity activity of *H. erinaceus* might not be specific to a menopausal model; it may be exerted on various subjects.

In study 1, the FBD group showed a significant reduction in white adipose tissue in the visceral and subcutaneous fat, while there was no significant difference in organ weight compared with the CD group. The reason why the FBD did not decrease body weight despite the reduction in

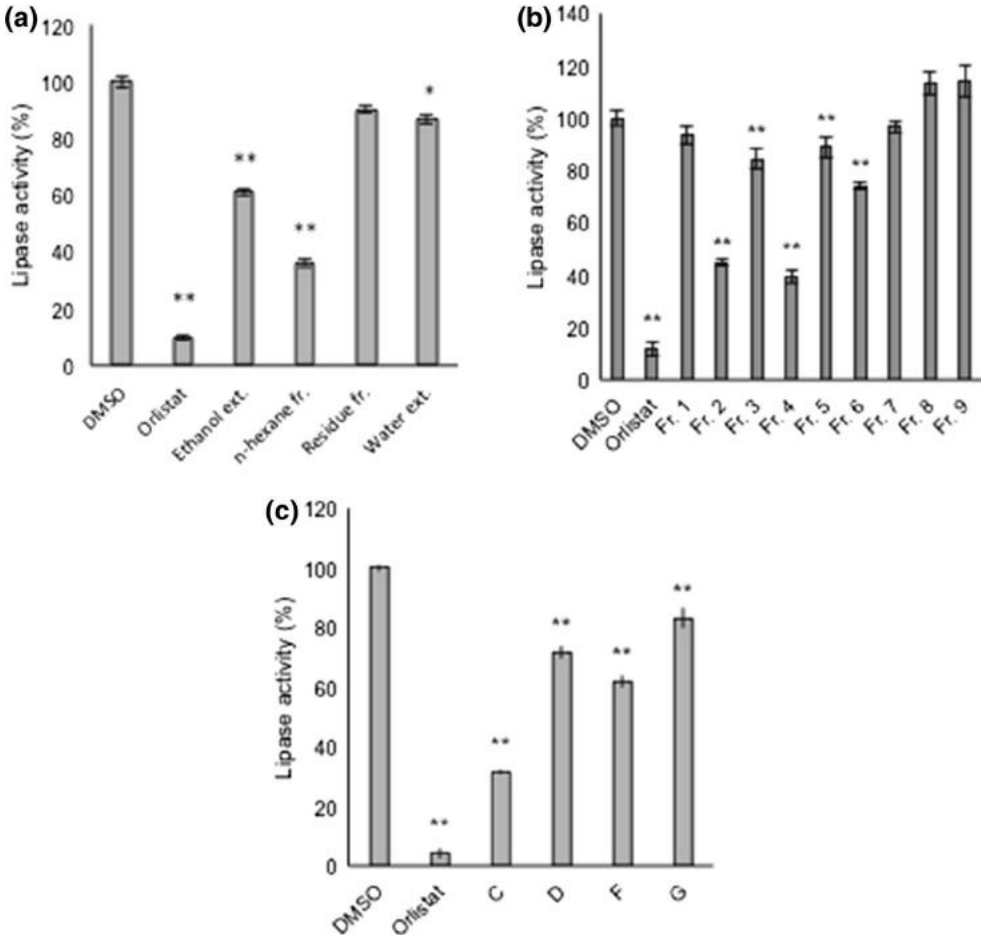


**Fig. 4** The effects of extracts, residue, and fractions (**a** and **b**) as well as isolated compounds (**c**) on lipase activity. DMSO was the negative control, orlistat was the positive control (1 μM). ** and * indicate significant differences in comparison with the negative control ($p < 0.01$ and $p < 0.05$, respectively, Dunnett $t$ test). $n = 4$. Mean ± SD. **a** The concentrations of the samples were all 400 μg/ml. **b** The concentrations of the samples were all 200 μg/ml. **c** *C*, *D*, *F*, and *G* refer to hericenone C, hericenone D, hericenone F, and hericenone G, respectively. The concentrations of isolated compounds were 400 μg/ml

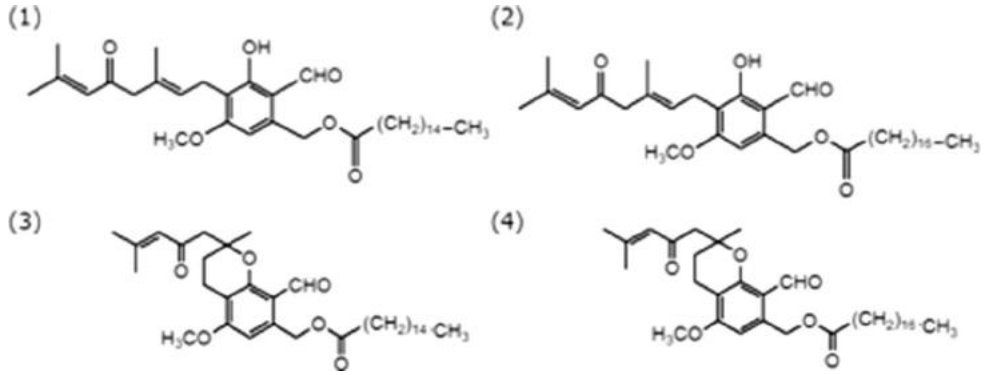


**Fig. 5** Structures of isolated compounds: **1** hericenone C; **2** hericenone D; **3** hericenone F; **4** hericenone G

white adipose tissue in visceral and subcutaneous fat is unclear, because we did not measure all organ weights. However, other factors aside from the amount of adipose tissue are also responsible for changes in body weight: muscle mass, amount of water in the body, and contents of the gut. Therefore, we believe that although the FBD decreases the amount of adipose tissue, it does not lead to a loss of body weight.

The amount of uterus tissue was not increased in the FBD group compared with the CD group. Estrogen and its receptor agonists increase the amount of uterus [27]; therefore, *H. erinaceus* did not play the role of estrogen or its receptor agonists. This result demonstrates that adverse effects such as those caused by hormone treatments can be avoided by using *H. erinaceus* instead, and that *H. erinaceus* can be taken by patients who cannot have hormone treatments.

This investigation showed that gain in body fat was inhibited by consuming the fruiting body of *H. erinaceus*. To clarify its active components, we divided the fruiting

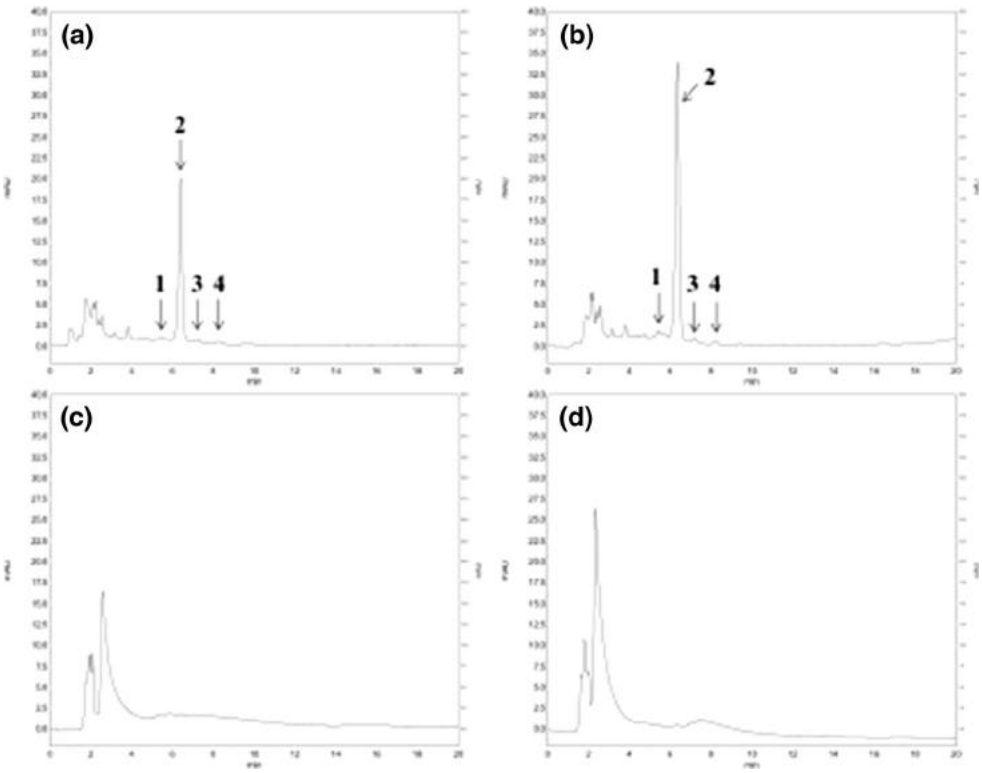


**Fig. 6** HPLC chromatograms of the **a** ethanol extract, **b** *n*-hexane fraction, **c** *n*-hexane residue fraction, and **d** water extract shown in Fig. 4a. The peaks are from: *1* hericenone C, *2* hericenone D, *3* hericenone F, and *4* hericenone G

body into an ethanol extract and the residue of the extract, and fed mice diets which contained each fraction. Both diets inhibited the increase in body fat. Thus, both the ethanol extract of *H. erinaceus* and the residue have inhibitory effects on fat gain.

To determine the mechanism of this effect, we measured the lipid levels in the matter excreted by mice from day 7 to day 14. The total amount of excreted matter was the same regardless of the diet; however, the FBD group showed increased lipids in its excreted matter. Additionally, there was no significant inhibitory effect in the EED or RD group, although the average lipid levels in their excreted matter increased; the increase in lipid excretion that occurred in the FBD group can therefore be attributed to components of both the ethanol extract and the residue. Accordingly, *H. erinaceus* has an inhibitory effect on lipid absorption; this is one of the mechanisms by which the increase in body fat is inhibited, and it is caused by components of both the ethanol extract and the residue.

Dietary fiber is a strong candidate for the active component in *H. erinaceus*. *H. erinaceus* is rich in dietary fiber (45.4% w/w). Dietary fiber interacts physically with lipids in the intestine (triacylglycerol, cholesterol, etc.), and this mixture is excreted without being absorbed [28]. Consequently, the residue, which contains a large amount of dietary fiber (Table S2 in the ESM), exerts a fat-reducing effect. On the other hand, the ethanol extract contained little dietary fiber. This suggests that there are other active components in *H. erinaceus* besides dietary fiber. We believe that lipase-inhibitory compounds may be the other component that contributes to the fat reduction effect.

Lipase is an essential enzyme that helps the body to absorb lipids. The most common form of lipid is triglycerides in the intestine, and this substance cannot be absorbed in the intestine. Lipase converts triglycerides into two fatty acid molecules and one 2-monoglyceride molecule; together with bile acid, these form micelles, which can be absorbed. Therefore, lipase-inhibitory compounds decrease lipid absorption and have been utilized in anti-obesity drugs such as orlistat [29].

As a consequence, in the present study, the ethanol extract inhibited lipase activity while the water extract made from the residue of the ethanol extract did not inhibit lipase activity. The in vitro results demonstrated that the high lipid excretion in the RD group was caused by dietary fiber, while that in the EED group was caused by a lipase-inhibitory effect.

We isolated the lipase-inhibitory compounds hericenone C, hericenone D, hericenone F, and hericenone G (compounds specific to *H. erinaceus*) in the ethanol extract by bioactive fractionation. The more hydrophilic the compound, the higher its lipase-inhibitory activity (hericenone C > hericenone F > hericenone D > hericenone G), which may be due to their enhanced ability to move in water. The ethanol extract and *n*-hexane fraction caused significant inhibition of lipase activity. The *n*-hexane fraction, which

contained the highest levels of hericenones, showed especially high lipase-inhibitory activity. However, the *n*-hexane residue fraction and the water extract, which did not contain hericenones, did not show inhibitory activity against pancreatic lipase. Therefore, it appears that the hericenones are responsible for the inhibition of pancreatic lipase activity. This is the first time that hericenone groups have been found to possess lipase-inhibitory activity. However, these compounds showed low activity, indicating that other inhibitory compounds are also present in the ethanol extract.

In this study, we investigated the effects of *H. erinaceus* on body fat in menopause. The results showed that *H. erinaceus* reduced body fat because the dietary fiber and lipase-inhibitory compounds—hericenone C, hericenone D, hericenone F, and hericenone G—present in this mushroom suppressed lipid absorption.

*H. erinaceus* has been shown to improve depression and anxiety in menopausal women in clinical practice [18], which we believe to be due—at least in part—to its fat-reducing effect, because obesity elevates the risk for depression and anxiety [30, 31].

In summary, the results of this investigation provide valuable insight into the potential benefits of *H. erinaceus* in menopause.

## Conclusions

Our results demonstrate that Yamabushitake (*Hericium erinaceus*) has anti-obesity activity in ovariectomized mice, and that this activity results from an ability to reduce lipid absorption. This reduction is caused by the dietary fiber and four compounds (hericenone C, hericenone D, hericenone F, and hericenone G) in Yamabushitake, which have lipase-inhibiting effects.

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**Compliance with ethical standards**

**Conflict of interest** The authors declare no conflict of interest.

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